Benchmarking Multi-View Tree-Level Oil Palm Bunch Counting Under a Fixed Detector

Muhammad Zainal Muttaqin[1], Fatma Indriani[1], Setyo Wahyu Saputro[1], Alia Rahmi[2], Dwi Kartini[1], Triando Hamonangan Saragih[1], Naufal Said[1] and Hartoni[3]
[1]Department of Computer Science, Faculty of Mathematics and Natural Sciences
[2]Department of Agro-Industrial Technology, Faculty of Agriculture
[3]Department of Agribusiness, Faculty of Agriculture
Lambung Mangkurat University, Banjarbaru, Indonesia
f.indriani@ulm.ac.id

***Abstract*. Oil palm harvest planning requires a per-tree count of bunches in each maturity class. Multi-view imaging reduces occlusion, but the same bunch reappears across photographs, so detections must be aggregated into unique counts. We compare five regression counters and eight feature sets on SawitMVC (953 trees, 3,992 images), using ground-truth (GT) detections and a fixed YOLO26m detector. The best observed Class ±1 accuracies are 98.05% and 77.48%, respectively; Tree ±1 accuracies are 92.20% and 32.62%. The class-level gap is 20.57 percentage points (95% CI 17.20–23.94), whereas no fixed-input counter pair differs significantly after multiplicity correction. Diagnostic matching finds 22.7% of test bunches undetected in every view, alongside substantial maturity confusion. The GT advantage persists in matched cross-validation, four YOLO26 checkpoints and a separately trained YOLO11m, whose matched-counter gap is 23.94 points. These results support prioritising detector output quality on this dataset. They do not establish a ceiling on counter improvement, and the diagnostic rules do not provide an additive decomposition of deployed error.**

***Keywords*. Oil palm, fresh fruit bunch, black bunch census, multi-view counting, object detection, YOLO, tree-level regression, agricultural computer vision.**

I. Introduction

Harvest planning in oil palm plantations depends on knowing how many bunches on each tree belong to each maturity class. The Black Bunch Census (BBC) records four classes, B1 to B4, with different ripening schedules [1]. Automating this survey could reduce manual effort and subjectivity, but requires the full class breakdown for each tree.

Bunches surround the trunk, so a single image misses fruit hidden by fronds, other bunches or the viewing angle [2]. Photographing each tree from four to eight sides improves coverage, but the same bunch can appear in several images. Summing detections therefore counts appearances separately (Fig. 1), although the census requires unique bunch counts.

Multi-view counting has been studied in other crops. Mango systems associate detections using navigation data and canopy geometry [3], while apple systems build maps from overlapping scans [4]. A comparative apple study identifies detection quality as a major source of counting error [5]. Other approaches use correspondence-free triangulation [6], few-view reconstruction [7], or neural fields [8]. Their dense overlap, moving sensors and calibrated geometry differ from our sparse views of individual trees.

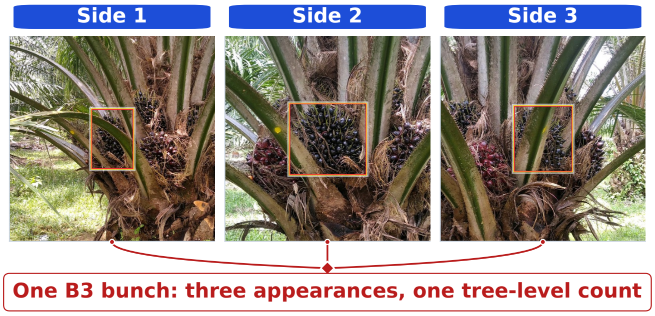

Fig. 1. One B3 bunch in three adjacent views: three image appearances must contribute one tree-level bunch count.

Regression offers an alternative by mapping detection statistics to a count vector [2], [4], [5], often with a YOLO detector providing the inputs [9], [10]. Evaluation only on detector outputs combines errors from both stages. Comparing the same counters with accurate annotations helps assess their dependence on input quality.

Oil palm detection remains difficult for occluded bunches and visually ambiguous maturity classes [11], despite work on two-stage and YOLO models [12]–[15]. A public video dataset provides frame-level annotations, but no tree-level counting labels or reported counting evaluation [16]. We study tree-level counting on SawitMVC [17], comparing five counters and eight feature sets under ground-truth (GT) and fixed-detector inputs. Paired statistical comparisons, feature ablation and detection diagnostics assess the main results; detector, threshold, resampling and estate analyses examine their wider scope.

II. Methodology

A. Dataset and Task Definition

SawitMVC contains 953 trees and 3,992 images, with four maturity classes and 4 to 8 views per tree [17]. The fixed split is 716 training, 96 validation and 141 test trees. The target is $\mathbf{y}_i = [y_{i,B1}, \ldots, y_{i,B4}]$; predictions $\hat{\mathbf{y}}_i$ are evaluated per tree. If $a_{i,c}$ counts class-c appearances across views, naive

TABLE I
DETECTOR PERFORMANCE AND APPEARANCE-LEVEL ERRORS.

	Validation		Test split, appearance level				
Class	mAP50	Recall	GT app.	Missed	Wrong	FP	Recall
B1	**0.746**	**0.801**	252	47	33	58	**0.683**
B2	0.425	0.433	496	161	209	158	0.254
B3	0.550	0.656	1409	494	65	393	0.603
B4	0.363	0.389	455	262	89	97	0.229
All	0.521	0.570	2612	964	396	706	0.479

Note: Missed/Wrong use GT classes; FP uses predicted classes. Test All recall is pooled; validation All is macro-averaged.

counting sets $\hat{y}_{i,c} = a_{i,c}$ and counts repeated appearances as separate bunches. Dataset construction, annotations and class composition are documented in [17].

B. Detection Conditions

Counters are fitted separately to features from GT boxes and classes and from cached YOLO26m outputs. This comparison measures aggregation with accurate inputs alongside the deployed pipeline. GT supplies an empirical reference; it is not a mathematical upper bound on future counting methods. The fixed YOLO26m detector uses 716 training trees, 60 epochs, batch 32, image size 640, patience 60 and seed 42 (Ultralytics 8.4.49). Inference uses confidence 0.25 and standard post-processing. This checkpoint is fixed throughout Sections III-A–III-C.

Sensitivity checks include YOLO26n/s/m checkpoints trained under an earlier 763/95/95 split. Counters are fitted on the 590 trees in both training splits and evaluated on 64 trees outside gradient training in both, which include detector validation trees. A separate YOLO11m run uses the official 716/96/141 split, the same epochs, batch, image size, patience and seed, 12 data-loader workers, and Ultralytics 8.4.140. Its checkpoint is selected on validation and frozen; Ridge+F_0 is fitted on 716 trees per input source. Software and architecture defaults differ, preventing attribution to architecture alone.

Table I reports validation mAP50 under the COCO convention [18] and test matching (Section III-D). Missed, Wrong and FP denote unmatched annotations, matched boxes with incorrect classes, and unmatched detections. Test recall requires both localisation and class correctness; its protocol differs from validation.

C. Feature Vectors

The baseline $F_0 = [s_{B1:B4}, m_{B1:B4}, \mu_{B1:B4}, n_v]$ has 13 dimensions: class totals, per-view maxima and means, and view count. For class c and view v, let $k_{c,v}$ be the count, P_c the confidence multiset, $s_c = \sum_v k_{c,v}$, $\mu_c = s_c/n_v$, and σ_c the population SD over all views, including empty ones. Ratio denominators use $\varepsilon = 10^{-6}$. Table II defines confidence (20), spatial (8), distribution (20) and composition (6) additions; their union with F_0 gives F_{all} (67). Eight sets are listed in Table IV. Feature ablation uses fixed inputs because GT confidences are all one.

TABLE II
DEFINITIONS OF THE TREE-LEVEL FEATURES.

Group (dim)	Entries	Definition
F_0 (13)	s_c, m_c, μ_c, n_v	total, per-view maximum, and per-view mean detections of class c; n_v is the number of views, 4 or 8 (global)
conf (20)	ΣP_c, $\overline{P}_c$, $\max P_c$, h_c, h_c^+	sum, mean, and maximum confidence; counts with $p \geq 0.5$ and $p \geq 0.6$
spatial (8)	$\overline{cy}_c$, $\overline{A}_c$	mean $cy = (y_{\min} + y_{\max})/(2H)$ and mean $A = wh/(WH)$, for image width W and height H
distrib (20)	σ_c, $\min_v k_{c,v}$, $\sigma_c/(\mu_c{+}\varepsilon)$, $\lvert\{v : k_{c,v}{>}0\}\rvert$, $1/(1{+}\sigma_c)$	SD and minimum across views, coefficient of variation, views with detections, and view-to-view consistency
compos. (6)	T, $s_c/(T{+}\varepsilon)$, $s_{B3}/(s_{B2}{+}s_{B3}{+}\varepsilon)$	all detections $T = \sum_c s_c$ (global), class shares, and the B3 share of the B2/B3 pair (global)

Note: Entries repeat for B1–B4 unless marked global. Empty confidence mean/max and area are 0; empty mean vertical position is 0.5.

D. Counter Models and Evaluation Metrics

Two heuristic baselines provide reference points. The naive sum counts all appearances, while a global divisor corrects for average repeated visibility. The divisor is the ratio of input appearances to unique GT bunches in the 716 training trees, fitted separately for each condition: $k = 1.891$ (GT) and 1.793 (fixed). No validation or test trees enter fitting.

Five regressors map each tree feature vector to its four class counts: linear regression, Ridge [19], ElasticNet [20], SVM [21] and Random Forest (RF) [22]. All use seed 42; continuous predictions are rounded to the nearest non-negative integer before scoring. Features are standardised except for RF. RidgeCV tests $\alpha \in \{0.01, 0.1, 1, 10, 100, 500\}$; MultiTaskElasticNetCV uses five folds and 3000 iterations. Multi-output RBF SVR uses three-fold GridSearchCV with $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$. RF uses 200 trees of maximum depth 10.

These settings preserve the released benchmark. Ridge and ElasticNet scale training data before internal CV; SVM scales within its CV pipeline. Outer test trees never enter fitting.

Class ± 1 accuracy measures correctness separately for each maturity class. Tree ± 1 is stricter: all four class counts must be within one bunch of the truth simultaneously. For N test trees, define $e_{i,c} = \hat{y}_{i,c} - y_{i,c}$ and $I_{i,c} = \mathbf{1}(|e_{i,c}| \leq 1)$. The accuracy measures are

$$\text{Class } \pm 1 = \frac{1}{4N} \sum_{i,c} I_{i,c}, \tag{1}$$

$$\text{Tree } \pm 1 = \frac{1}{N} \sum_{i} \mathbf{1}\left(\sum_{c} I_{i,c} = 4 \right). \tag{2}$$

Macro MAE averages $|e_{i,c}|$ over trees and classes; macro RMSE averages the four class RMSEs, emphasising large errors. Class bias is mean $e_{i,c}$ (positive: overcount; negative: undercount). Total-count MAE averages $|\sum_c e_{i,c}|$; total ±1 is the fraction with $|\sum_c e_{i,c}| \leq 1$. These inventory measures allow class errors to cancel, unlike Tree ±1. Directional and absolute errors matter for yield estimates, but per-tree metrics do not alone determine block-level forecast error. Code and results are available.[1]

E. Uncertainty and Validation Protocol

Bootstrap confidence intervals resample trees 10,000 times (seed 42), reusing the same indices for paired comparisons. Intervals cover per-class and aggregate metrics. Exact paired permutation tests exchange model outcomes within trees, preserving dependence among their four classes. Holm adjustment covers ten counter pairs within each condition; seven Ridge feature comparisons against F_0 form a separate family. We report raw p and adjusted p_H separately. Wilcoxon tests on per-tree macro absolute error and exact McNemar tests on Tree ±1 address different endpoints. These analyses condition on fitted models, excluding configuration selection and detector retraining; nonsignificance does not establish equivalence.

"Best" denotes the highest observed test score among compared configurations.

A separate selection check ranks all 40 model/feature candidates in each condition on the 96 validation trees: highest Class ±1, then lowest macro MAE, fewer dimensions, and model/feature name as deterministic tie-breaks. Only training trees enter model fitting; test labels do not select the winner. This reanalysis uses the historical test set, not newly collected test data.

Repeated five-fold cross-validation (CV), with five repeats, jointly stratifies by estate and dominant class. GT and fixed Ridge+F_0 use identical folds over 237 trees outside detector gradient training (189–190 fit trees per fold). This pool includes 96 detector validation trees, so a stricter check uses the 141 original test trees (112–113 fit trees per fold). The detector stays frozen. Fold SDs are descriptive because repeated folds overlap. Estate transfer evaluates counters; the detector was trained on both estates.

III. Results and Discussion

A. Counting Under the Two Detection Conditions

Table III shows why repeated appearances need to be accounted for even with accurate detections. Naive summation performs poorly, whereas the calibrated divisor recovers most of the loss and compact regressors improve further. The four non-RF counters span only 0.53 percentage points (pp) on Class ±1 and 2.13 pp on Tree ±1. Their pairwise Class ±1 tests give $p_H = 1$. ElasticNet exceeds RF by 2.13 pp (95% CI 0.89–3.55; $p = 0.0042$, $p_H = 0.0376$); SVM also exceeds RF ($p_H = 0.0342$).

The result changes substantially with detector inputs. The strongest F_0 Class ±1 scores have 95% CIs of 97.0–99.1% under GT and 73.2–79.6% under the fixed detector. ElasticNet loses 21.63 pp between conditions (CI 18.44–25.00; $p < 0.001$), while the five fixed-input counters span only 3.19 pp. None of their ten comparisons is significant; the largest, ElasticNet versus RF, gives CI 0.00–6.38, $p = 0.0628$ and $p_H = 0.628$. The much larger difference between input conditions supports investigating detector outputs, although this test set cannot establish counter equivalence.

The class pattern is consistent across counters: fixed-input accuracies are B1 95.74–96.45%, B2 73.76–80.14%, B3 55.32–56.03%, and B4 67.38–73.05%. B3 has the largest counting loss despite higher test recall than B2 and B4. Recall alone therefore does not rank counting difficulty: larger counts and class confusion also affect the absolute ±1 criterion.

TABLE III
Counting performance with baseline features.

Method	GT detections			Fixed detector		
	Class ±1	Tree ±1	MAE	Class ±1	Tree ±1	MAE
Naive sum	50.00	6.38	2.142	50.71	10.64	2.447
Global divisor	95.57	86.52	0.356	70.21	22.70	1.188
ElasticNet	**98.05**	**92.20**	0.277	**76.42**	29.79	**1.043**
SVM	97.87	91.49	**0.266**	74.82	29.08	**1.043**
Ridge	97.70	90.78	0.275	76.06	28.37	1.053
Linear reg.	97.52	90.07	0.277	75.71	**30.50**	1.048
Random forest	95.92	84.40	0.365	73.23	26.95	1.110

Note: Accuracy is in %; MAE is macro MAE. The divisor is fitted separately per input condition using training trees only.

TABLE IV
Ridge feature ablation under detector inputs.

Features	Dim	Class ±1	Tree ±1	MAE	CV Class ±1
F_0	13	76.06	28.37	1.053	72.70 ± 2.35
F_0+C	33	75.89	29.79	1.059	**73.26** ± 2.36
F_0+S	21	76.60	30.50	1.046	72.65 ± 2.37
F_0+D	33	74.82	28.37	1.060	72.89 ± 1.96
F_0+C+S	41	76.24	29.08	1.059	72.40 ± 2.46
F_0+C+D	53	76.42	29.79	1.060	72.59 ± 2.02
F_0+D+S	41	75.71	30.50	1.048	72.34 ± 2.33
F_{all}	67	**77.48**	**32.62**	**1.035**	72.34 ± 2.08

Note: C: confidence; S: spatial; D: distribution. CV: mean ± fold SD on 237 trees, including detector validation trees. MAE is macro MAE.

B. Feature Ablation

To test whether richer features recover the gap, we fix Ridge and vary its inputs (Table IV). Class ±1 ranges from 74.82% (F_0+D) to 77.48% (F_{all}). The full set gains 1.42 points at class level and 4.25 at tree level over F_0. The CV column applies the protocol in Section II-E to the same eight sets.

The class-level gain is not statistically resolved (95% CI -0.53 to $+3.37$; $p = 0.201$, $p_H = 1$). CV means span only 0.92 points, less than half their fold SDs, and F_{all} no longer leads. No Class ±1 comparison with F_0 is significant after adjustment. Thus evidence for these particular additions is limited; other representations may still recover useful information.

[1]Source code: https://github.com/ULM-SawitMVC/Baseline-SawitMVC.

TABLE V
PER-CLASS COUNTING PERFORMANCE.

Configuration	Metric	B1	B2	B3	B4	All
GT,	Acc. ±1 (%)	100.00	99.29	93.62	99.29	98.05
ElasticNet+F_0	MAE	0.050	0.227	0.603	0.227	0.277
	RMSE	0.223	0.519	0.855	0.491	0.522
	Bias	$-.050$	$+.043$	$-.064$	$-.028$	$-.099$
Fixed,	Acc. ±1 (%)	97.16	82.98	57.45	72.34	77.48
Ridge+F_{all}	MAE	0.369	1.014	1.553	1.206	1.035
	RMSE	0.652	1.437	1.984	1.598	1.418
	Bias	$+.014$	$-.078$	$-.177$	$+.071$	$-.170$

Note: All is the class mean, except Bias, which is total signed error. Counts and errors are in bunches per tree.

C. GT vs. Fixed-Detector Gap

The best observed configurations (Table V) differ by 20.57 pp on Class ±1 (95% CI 17.20–23.94; $p < 0.001$) and 59.57 pp on Tree ±1. Class losses are B1 2.84, B2 16.31, B3 36.17 and B4 26.95 pp. Using Ridge+F_0 in both conditions retains a 21.63-point gap (CI 18.44–25.00; $p < 0.001$), so different counter families do not explain it. Macro RMSE rises from 0.522 to 1.418; total-count MAE from 0.979 to 1.787; total ±1 falls from 78.01% to 48.94%. Block-level forecast error also depends on cancellation and dependence across trees.

Validation selects Ridge+F_0 (GT) and SVM+$(F_0 + \text{conf})$ (fixed): test Class ±1 is 97.70% versus 74.11%, a 23.58-point gap (CI 20.21–27.13; $p < 0.001$); Tree ±1 is 90.78% versus 27.66%. The gap persists without test-based selection.

Signed bias adds direction to the absolute errors. Fixed Ridge+F_{all} underestimates B2 and B3 and overestimates B4 despite low recall; its total bias is -0.170 bunches/tree. Bias combines missed detections, false positives and class confusion, so it cannot be assigned to one confusion direction.

D. Detection and Aggregation Diagnostics

Two mechanisms are candidates for the residual: bunches missed in every view and maturity-class confusion. A counter may infer missing counts statistically, but cannot directly associate a bunch with no matched detection. To inspect these mechanisms, every detection on the test trees is matched against the annotations, and each matched detection inherits the identity of the ground-truth bunch it hit. Matching is greedy in order of confidence and class-agnostic, and accepts a pair at IoU ≥ 0.5 in pixel coordinates, so classification errors among matched boxes are separated from unmatched boxes. An unmatched annotation can reflect absence, IoU below threshold, or matching competition; it is not a pure measure of visibility or localisation failure.

Table I reports the matching results. Of 2,612 test appearances, 964 (36.9%) are unmatched and 396 (15.2%) are localised but assigned another maturity class, leaving 1,252 (47.9%) both found and labelled correctly; in the other direction, 706 of 2,354 detections (30.0%) match no annotation, at a mean confidence of 0.337, well below the 0.487 of matched detections. The failure modes are class-specific: B2 loses more appearances to class confusion (209, of which 158 are read as B3) than to missed detection (161), whereas B4 is dominated by unmatched annotations (262 of 455). Counted as physical bunches, 317 of the 1,397 test bunches (22.7%) are never detected in any view, from 8.5% for B1 to 42.3% for B4.

Figure 2(a) places three identity-based diagnostic rules alongside the GT counter, deployed counter and naive sum. Counting each matched physical bunch once with its GT class gives 86.17% Class ±1 Acc. Replacing that GT class by a confidence-weighted vote across matched detections gives 71.81%; adding every unmatched detection as a separate bunch gives 62.23%. Identity is known only for matched detections, so the last rule does not deduplicate false positives across views.

The losses of 13.83 points from exact GT counts to the first rule, 14.36 between the first two rules, and 9.58 between the last two depend on their ordering, voting and treatment of unmatched detections. They cannot be added to the GT counter's 1.95-point error to explain deployment. The deployed 77.48% exceeds two diagnostic rules because learned aggregation can compensate statistically. Likewise, the 8.69-point difference between 86.17% and deployment is not an association bound: the first rule uses GT classes and excludes unmatched boxes. These diagnostics expose detection failures, while the matched-input comparison measures their combined effect on a specified counter. Isolating association error requires an explicit module and controlled interventions on it.

E. Sensitivity and Resampling

A conclusion drawn from one detector and one split needs additional checks. Table VI compares detector families, checkpoints and thresholds. With the same Ridge+F_0 protocol on 141 test trees, YOLO11m differs from YOLO26m by -2.30 pp on Class ±1 (95% CI -4.96 to $+0.35$; $p = 0.118$). GT retains a 23.94-point advantage over YOLO11m (CI 20.57–27.30; $p < 0.001$). YOLO11m has 976 unmatched and 399 misclassified GT appearances out of 2,612, and 463 unmatched detections out of 2,099. This supports the GT advantage in another family, without establishing equivalence or a general architecture ranking.

The threshold sweep filters a second inference pass at confidence 0.05, refitting Ridge+F_{all} on 716 trees at each of eleven thresholds. At 0.25, this cache has 2,353 detections versus 2,354 originally, with the same counting accuracies. Threshold 0.10 ties the best Class ±1 (77.48%) and raises Tree ±1 to 38.30% from 32.62%; 0.70 reduces Class ±1 to 63.12%, while 0.05 yields 76.24% from 8,807 detections. Across eleven thresholds, none exceeds the reported best Class ±1, although the whole-tree endpoint can improve. This exploratory test sweep does not independently select a new threshold.

On the 64-tree subset, four YOLO26 checkpoints span 70.70–73.83% (3.13 pp), versus GT 97.27%, a 23.44-point gap to the best checkpoint. Capacity and training splits are confounded; the subset includes detector validation trees.

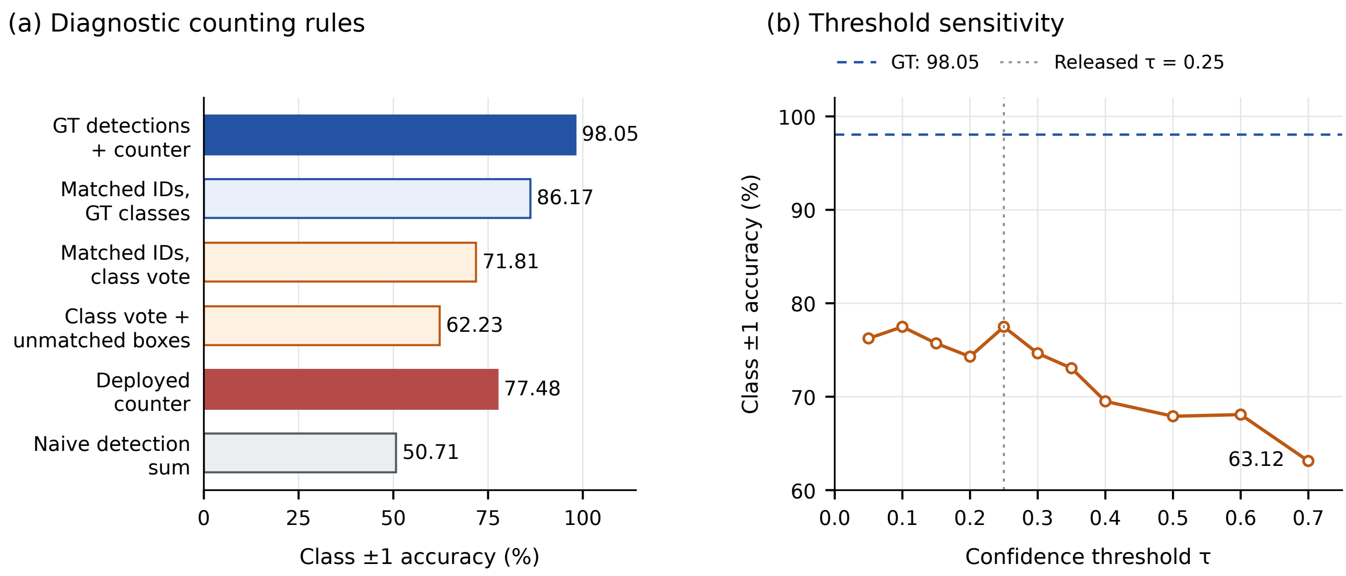

Fig. 2. Detection and aggregation diagnostics: (a) six counting rules; (b) confidence-threshold sensitivity of Ridge+F_{all}. Diagnostic rules are separate estimators, not additive error components.

TABLE VI
DETECTOR FAMILY, CHECKPOINT AND THRESHOLD SENSITIVITY.

Setting	Detections	Recall	Class ±1	Tree ±1
Family, Ridge+F_0, 141 test trees				
YOLO11m	2099	0.445	73.76	26.24
YOLO26m	2354	0.442	76.06	28.37
GT detections	2612	1.000	97.70	90.78
Checkpoint, Ridge+F_0, 64 evaluation trees				
YOLO26n	1013	0.442	72.66	34.38
YOLO26s	825	0.383	72.27	26.56
YOLO26m	950	0.427	70.70	28.12
YOLO26m, new split	1067	0.442	**73.83**	31.25
GT detections	1203	1.000	97.27	89.06
Operating point, Ridge+F_{all}, 141 test trees				
$\tau = 0.05$	8807	0.591	76.24	34.75
$\tau = 0.10$	5522	0.556	**77.48**	**38.30**
$\tau = 0.25$	2353	0.441	**77.48**	32.62
$\tau = 0.35$	1488	0.355	73.05	23.40
$\tau = 0.50$	799	0.245	67.91	21.28
$\tau = 0.70$	125	0.086	63.12	14.89

Note: Recall is macro class-correct appearance recall at IoU ≥ 0.5; threshold recall is recomputed after filtering. GT counts are annotated appearances; their recall is 1 by construction. Checkpoint rows use the earlier split unless marked.

Matched 5×5 CV gives Class ±1 means and fold SDs of $97.72 \pm 1.02\%$ (GT) and $72.70 \pm 2.35\%$ (fixed) on 237 trees (gap 25.03 pp); the 141-test-tree check gives $97.16 \pm 1.23\%$ and $74.08 \pm 2.96\%$ (gap 23.08 pp). Both analyses support the GT advantage when the detector is frozen. However, their smaller counter training sets prevent interpreting the original gap as conservative.

For Ridge+F_{all}, transfer from 641 DAMIMAS training trees to 24 LONSUM evaluation trees gives 71.88%, versus 72.92% when trained on 75 LONSUM trees. Reverse transfer to 213 DAMIMAS evaluation trees gives 68.78%. Estate and training size are confounded; the detector saw both estates.

To examine sensitivity of the vertical spatial features, we transform test features using $\overline{cy} \mapsto \text{clip}(a\overline{cy} + b, 0, 1)$ over 35 settings of $a \in [0.90, 1.10]$ and $b \in [-0.10, 0.10]$; the clip leaves missing-class sentinels at 0.5. Training uses unperturbed data. Ridge+F_{all} loses at most 1.42 pp; Gaussian noise with $\sigma = 0.05$ and seed 42 costs 0.53 pp. These perturbations test sensitivity to feature shifts. Physical camera pitch also changes perspective, box area, visibility and detector outputs, which this experiment does not reproduce. Robustness to actual camera rotation therefore remains unverified.

F. Why Random Forest Trails

Under GT inputs, RF trails ElasticNet and SVM after Holm correction. Each tree averages training targets within a leaf, and the forest averages those predictions. It therefore cannot extrapolate beyond the training count range. When high-count examples are sparse, averaging can pull estimates towards more common counts. Slopes from regressing predictions on true counts are 0.754–0.888 for RF versus 0.865–0.951 for linear counters; relative prediction SD is smaller for RF in every class. B1 predictions never exceed three bunches, although the maximum is five in training and six in the test set; MAE is 0.149 versus Ridge's 0.050. RF trails ElasticNet by 3.07 pp in the highest-load bin (94.74% versus 97.81%). These diagnostics suggest shrinkage, without proving causality or excluding benefits from tuning. With fixed inputs, the ElasticNet versus RF difference remains unresolved ($p = 0.0628$, $p_H = 0.628$).

G. Limitations

Evidence covers one dataset from two South Kalimantan estates, the smaller with 99 trees. Generalisation to other regions, varieties and acquisition conditions remains uncertain; no plantation was excluded from detector training. YOLO11m has one training run, software versions and defaults differ, and the 64-tree analysis is descriptive. Counter CV does not measure detector training variability. The dataset lacks calibrated pitch measurements and repeated captures at known angles, so feature sensitivity does not establish physical pitch robustness. Test-based highlighting and reuse of the historical test set limit confirmatory claims. Counters ignore spatial and temporal regularities; tree resampling ignores dependence among neighbours. B2/B3 and B3/B4 are visually ambiguous, but no inter-annotator study quantifies their contribution to confusion.

IV. Conclusion

The GT advantage persists across counter choices, validation-based selection, matched resampling and an additional detector family. For BBC, the results support improving B3/B4 detection and B2-to-B3 classification alongside aggregation. Evaluating the same counter under GT and detector inputs helps practitioners assess input quality alongside aggregation. Class and total-count metrics distinguish maturity allocation from inventory error: a plausible total can conceal errors that affect the harvest schedule. Reporting signed bias alongside absolute error also helps assess systematic overcounting or undercounting. Cached predictions, fixed splits and code allow comparisons without repeating detector inference. These benefits are limited by the dataset's two unequal estates and untested acquisition changes; the results do not provide an association-error budget or a limit on what future counters can achieve. Future work should compare explicit association [6], [7], [23] with learned aggregation under matched inputs, broaden detector evaluation, repeat detector training, and test an independent plantation and measured camera-pitch changes before extending claims to operational harvest forecasts.

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